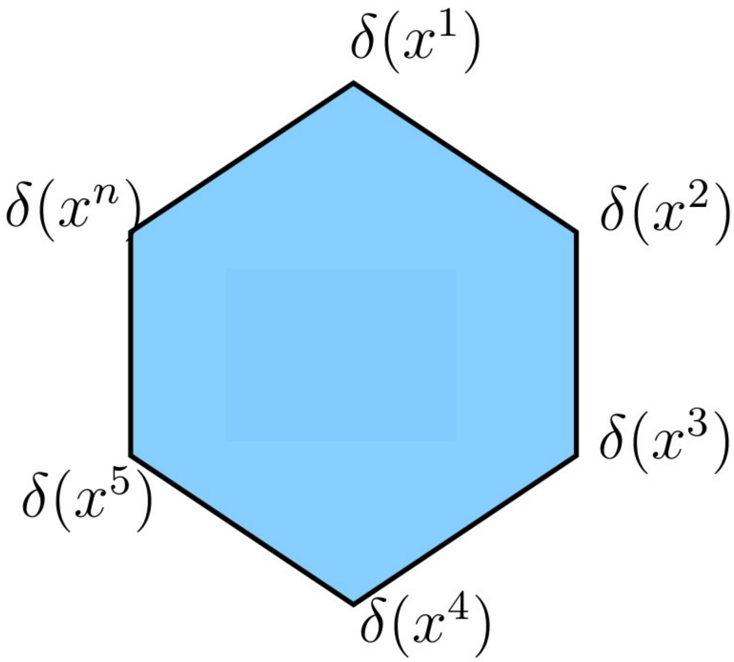
Marginal Polytope

Marginal polytope: $$M := \mathrm{conv}\{\delta(x) \in \mathbb{R}^I : x \in X_V\}$$



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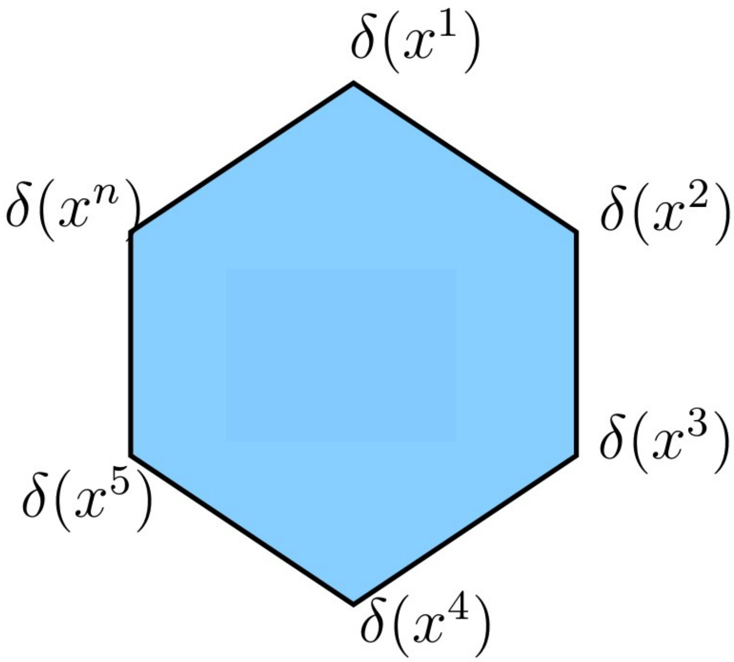
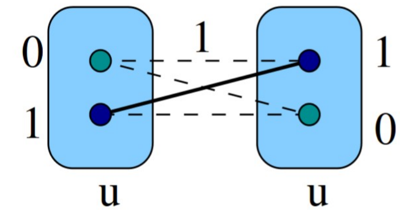
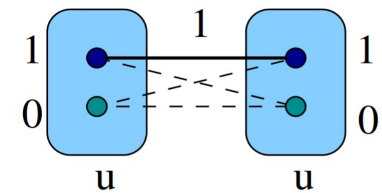


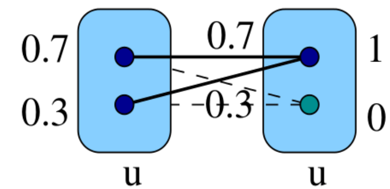


$$\delta(x^1) = \underbrace{01}_{u}\ \underbrace{0100}_{uv}\ \underbrace{10}_{v}$$



$$\delta(x^2) = \underbrace{10}_{u}\ \underbrace{1000}_{uv}\ \underbrace{10}_{v}$$

$0.3x^1 + 0.7x^2$:



$$0.3\delta(x^1) + 0.7\delta(x^2) = \underbrace{0.7\,0.3}_{u}\ \underbrace{0.7\,0.3\,00}_{uv}\ \underbrace{10}_{v}$$

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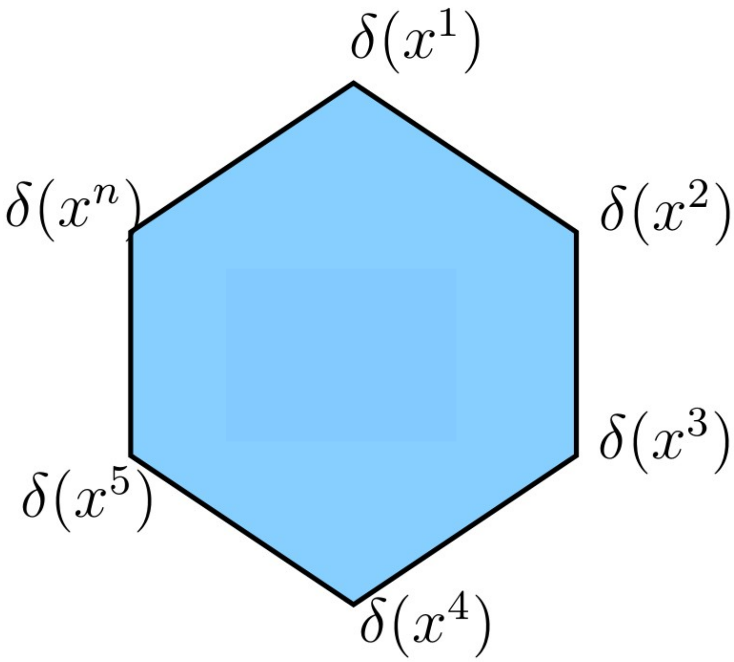
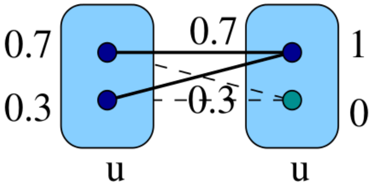


Corollary 1:

$$\mu_v \in \Delta^{X_v} \text{ for any } v \in V$$

$$\mu_{uv} \in \Delta^{X_{uv}} \text{ for any } uv \in \mathcal{E}$$

$0.3x^1 + 0.7x^2$:



$$0.3\delta(x^1) + 0.7\delta(x^2) = \underbrace{0.7\;0.3}_{u}\;\underbrace{0.7\;0.3\;0\;0}_{uv}\;\underbrace{1\;0}_{v}$$

Marginal Polytope

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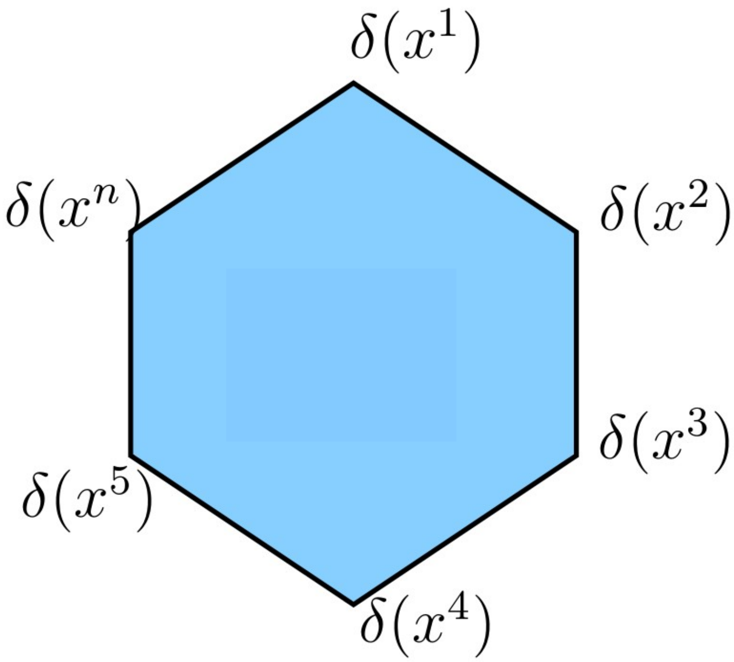

Proposition 2:

$$\min_{x \in X_V} \langle \theta, \delta(x) \rangle = \min_{\mu \in M} \langle \theta, \mu \rangle$$

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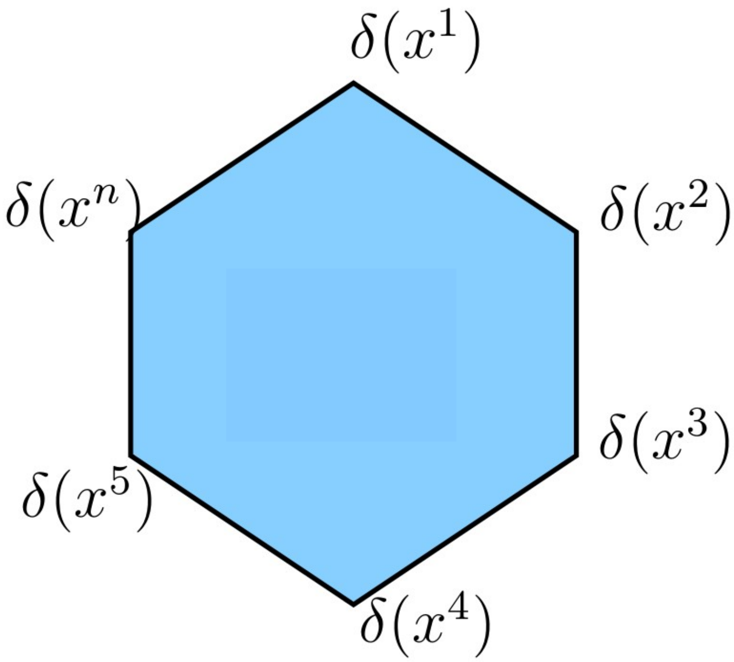

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NP-hard problem = linear problem?

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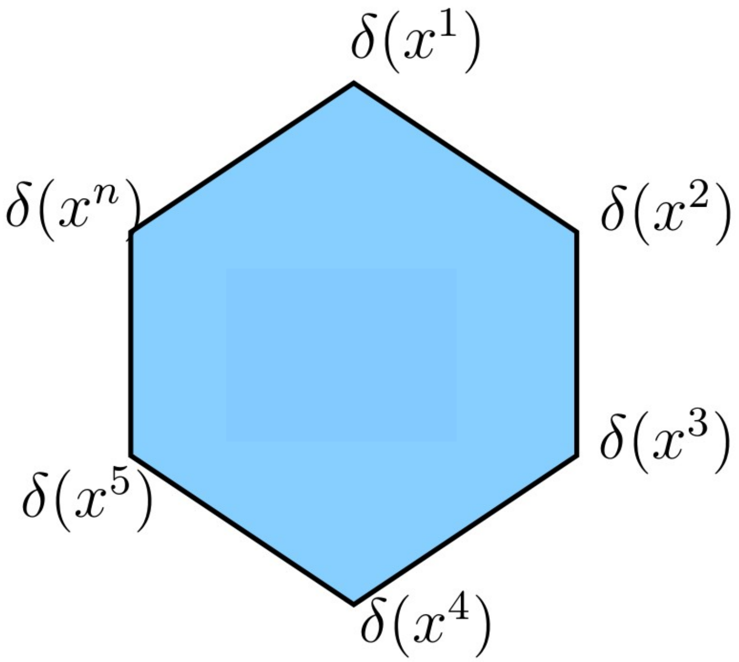

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NP-hard problem = linear problem?

$$\sum_{x \in X_V} p_x \delta(x) = \mu$$

$$\sum_{x \in X_V} p_x = 1$$

$$p_x \geq 0, \quad x \in X_V$$

Yes, with an exponential number of constraints!

Marginal Polytope

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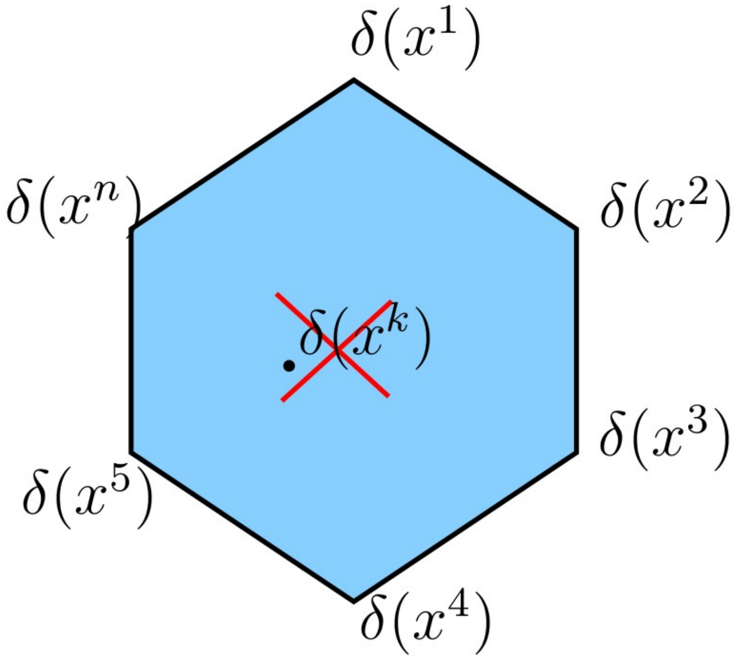

Proposition 3:

For any $x \in X_V$ the vector $\delta(x)$

is a vertex of M .

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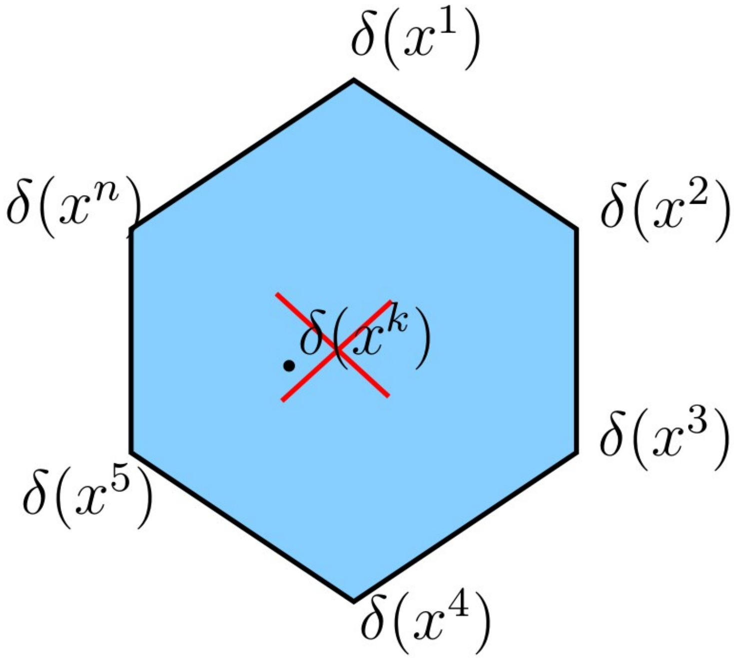

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$$\theta_w(s) = \begin{cases} 0, x_w = s \\ \infty, \text{otherwise} \end{cases}$$

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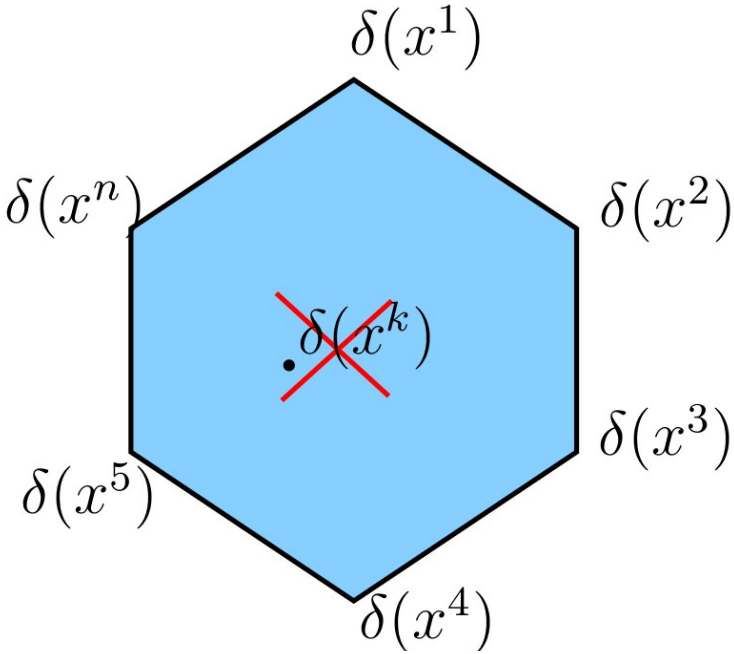

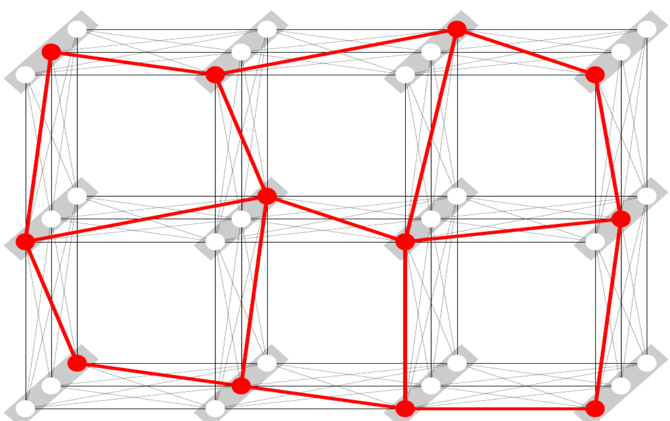

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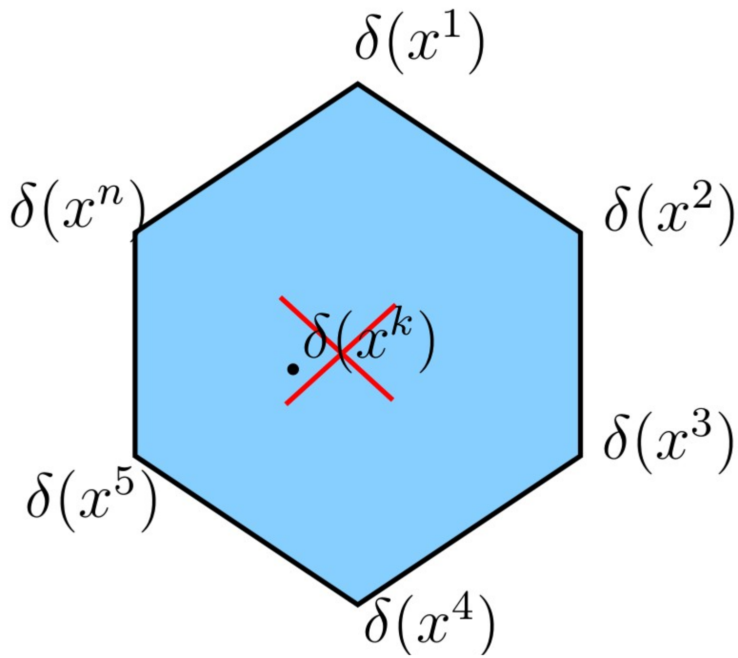

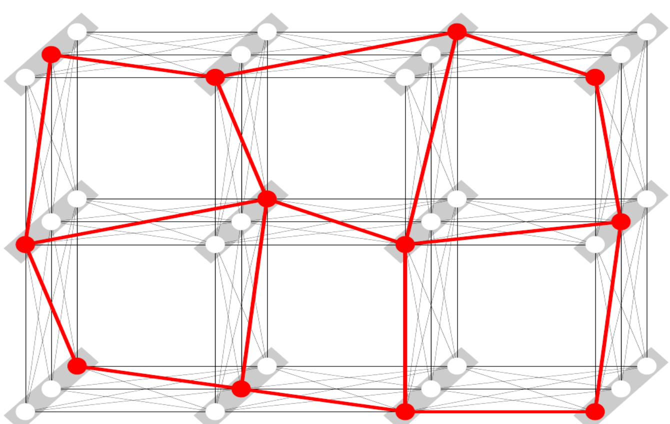

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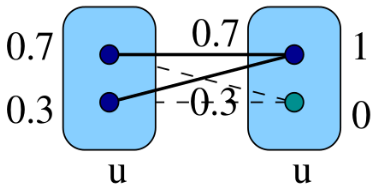
$0.3x^1 + 0.7x^2$:



$$0.3\delta(x^1) + 0.7\delta(x^2) = \underbrace{0.7\,0.3}_{u}\underbrace{0.7\,0.3\,0\,0}_{uv}\underbrace{1\,0}_{v}$$

$$\cancel{p_x \geq 0, \; x \in X_V}$$

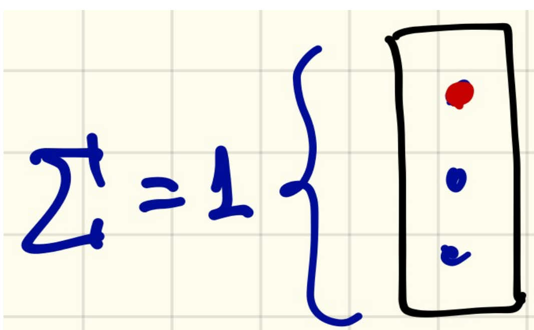
- non-local, exponential number

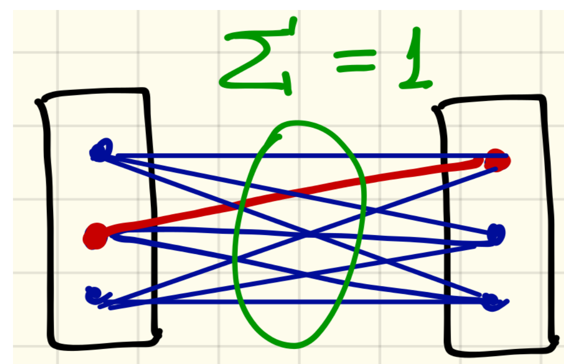
Local constraints are required to keep their number polynomial. Recall:

$$\sum_{s \in X_w} \mu_w(s) = 1, \qquad w \in V \times \mathcal{E}$$

- simplex constraints

$$\mu_w \geq 0\,.$$





Local (Marginal) Polytope

$$\cancel{p_x \geq 0,\ x \in X_V}$$ - non-local, exponential number

Local constraints are required to keep their number polynomial. Recall:

$$\sum_{s \in X_w} \mu_w(s) = 1, \qquad w \in V \times \mathcal{E}$$

$$\mu_w \geq 0\,.$$

- simplex constraints

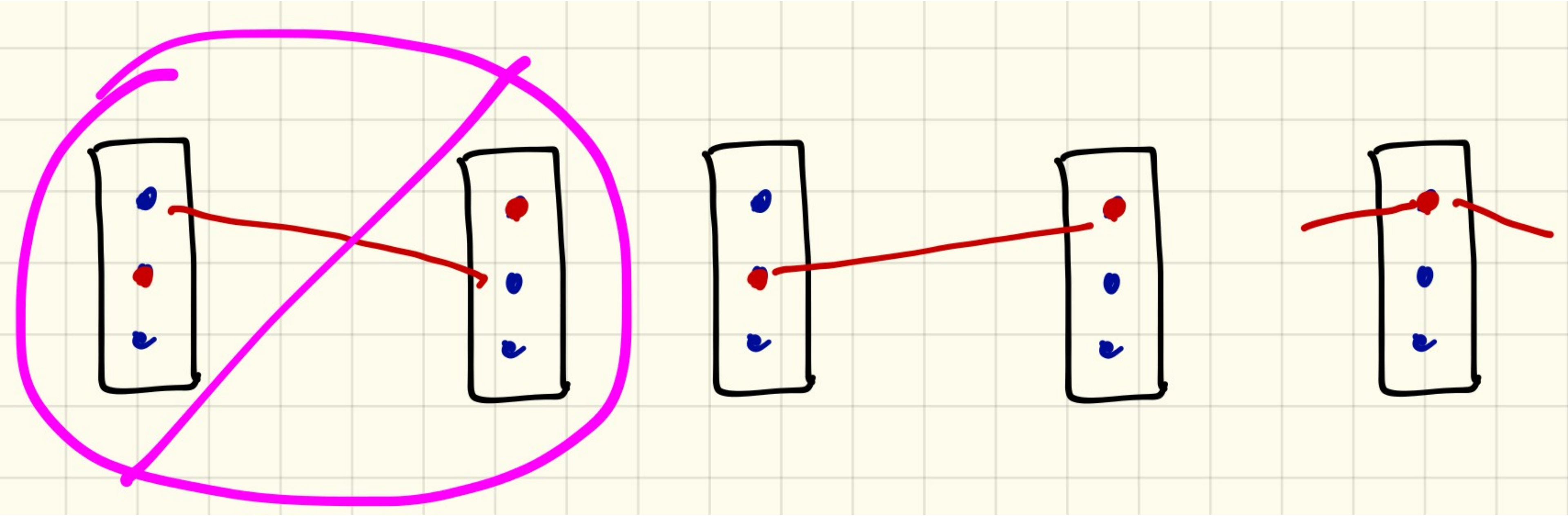
Insufficient, the inference problem separates out into independent parts for each factor:

$$\min_{\mu} \langle \theta, \mu \rangle$$
$$\text{s.t.} \ \sum_{s \in X_w} \mu_w(s) = 1,\ w \in V \times \mathcal{E}$$
$$\mu_w \geq 0\,.$$

$$\Rightarrow$$

$$\sum_{w \in V \times \mathcal{E}} \min_{\mu_w \in X_w} \langle \theta_w, \mu_w \rangle$$
$$\text{s.t.} \ \sum_{s \in X_w} \mu_w(s) = 1$$
$$\mu_w \geq 0\,.$$

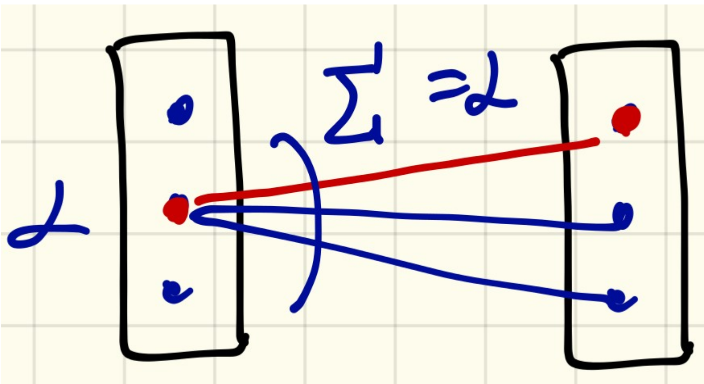
Local (Marginal) Polytope

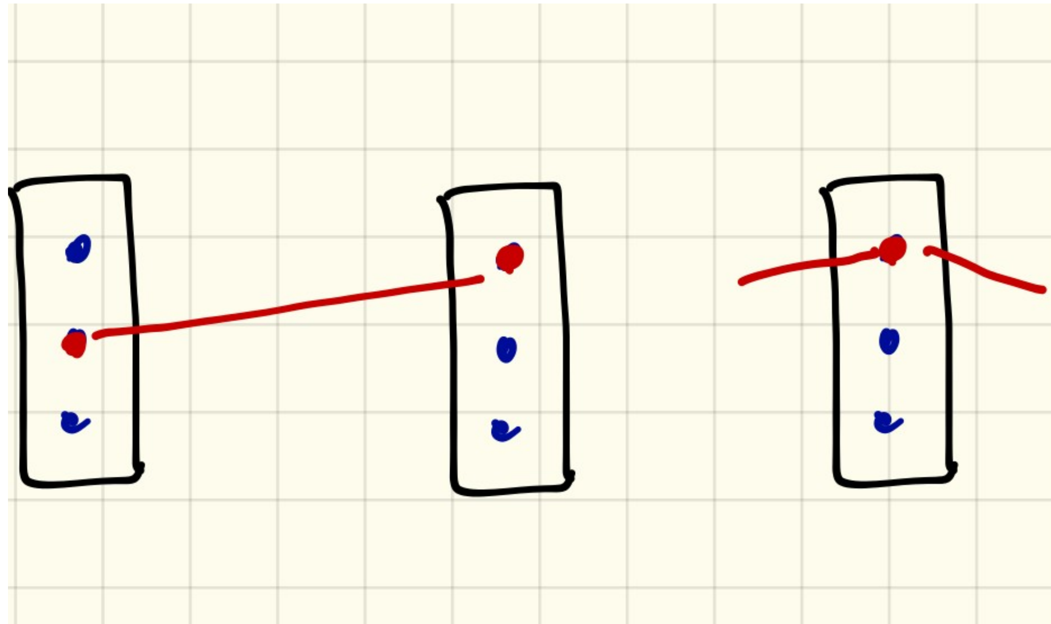


$$\mu_u(s) = 1 \;\Rightarrow\; \forall v \in N_b(u)\ \exists t \in X_v\colon \mu_{uv}(s,t) = 1$$

$$\mu_{uv}(s,t) = 1 \;\Rightarrow\; \mu_u(s) = 1 \text{ and } \mu_v(t) = 1(1)$$

Local (Marginal) Polytope





$$\sum_{s \in X_u} \mu_{uv}(s,t) = \mu_v(t),\ \forall\ v \in V,\ t \in X_v$$

$$\sum_{t \in X_v} \mu_{uv}(s,t) = \mu_u(s),\ \forall\ u \in V,\ s \in X_u\ .$$

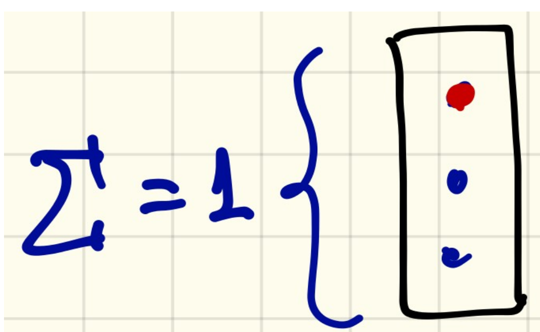
Local (Marginal) Polytope

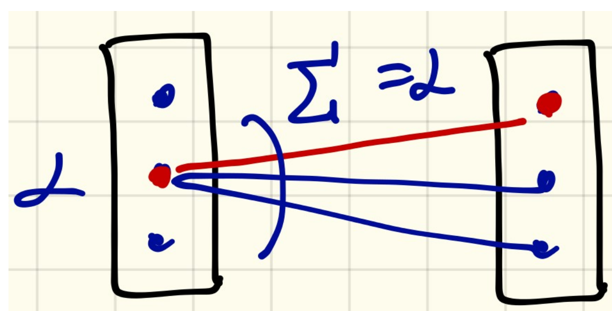
$$\mu^* = \arg\min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$$

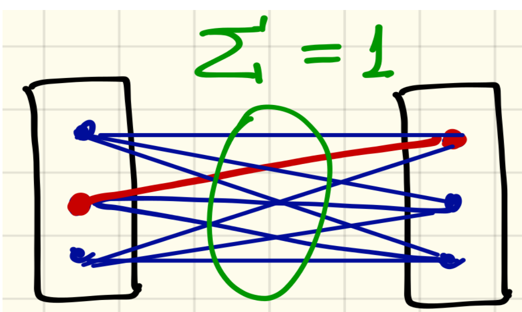
Local polytope (LP) relaxation; relaxed problem

μ - relaxed labeling $\quad$ μ^* - relaxed solution

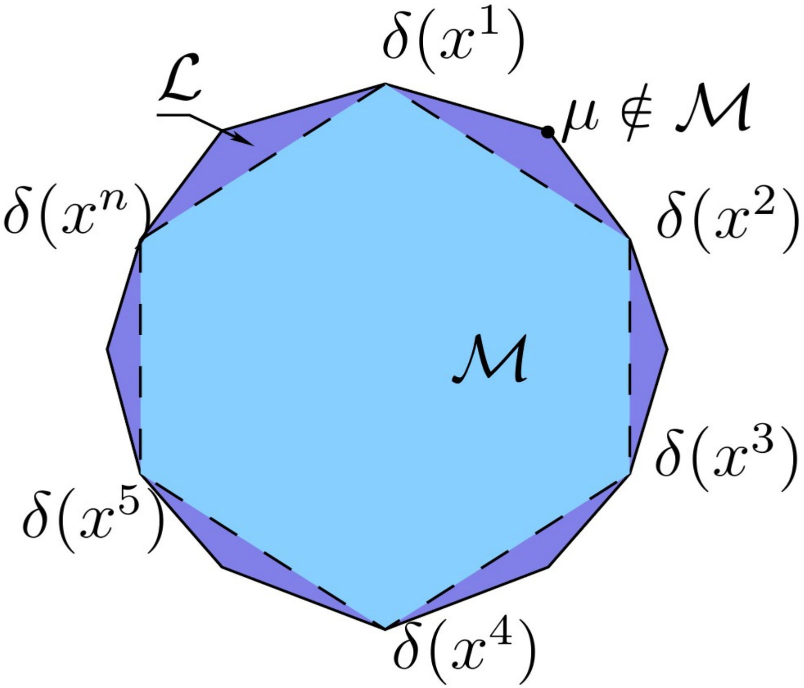
$$\mathcal{L} := \begin{cases} \sum_{s \in X_u} \mu_{uv}(s,t) = \mu_v(t), & v \in V,\ t \in X_v \qquad (a) \\ \sum_{t \in X_v} \mu_{uv}(s,t) = \mu_u(s), & u \in V,\ s \in X_u \qquad (b) \\ \sum_{t \in X_v} \mu_v(t) = 1, & v \in V \qquad (c) \\ \mu \geq 0, & \qquad (d) \\ \sum_{(s,t) \in X_{uv}} \mu_{uv}(s,t) = 1, & uv \in \mathcal{E},\ (s,t) \in X_{uv} \qquad (e) \end{cases}$$







Local (Marginal) Polytope

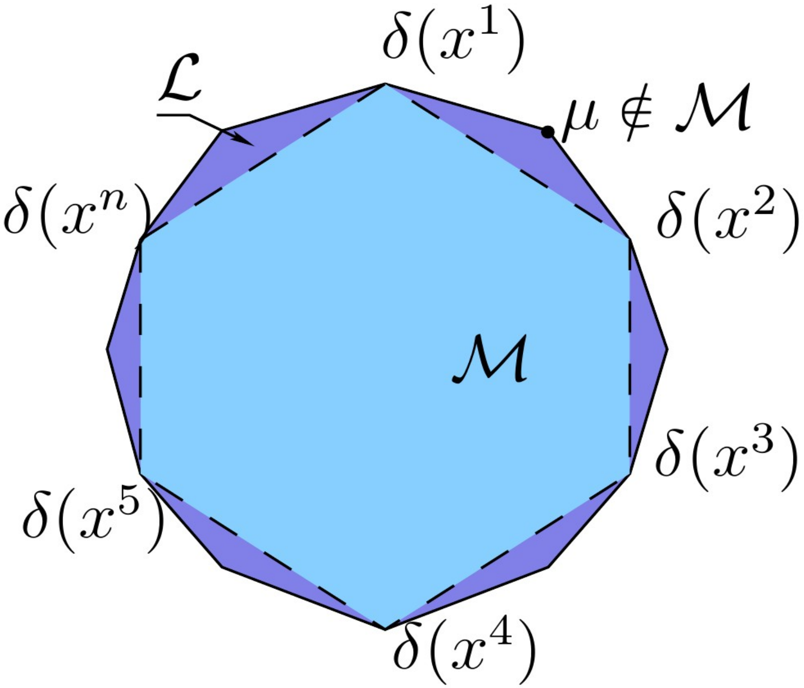


Proposition 4:

For any $x \in X_V$

$\delta(x)$ is a vertex of $\mathcal{L}$.

Proposition 5: $\mathcal{M} \subseteq \mathcal{L}$



Proposition 4:

For any $x \in X_V$

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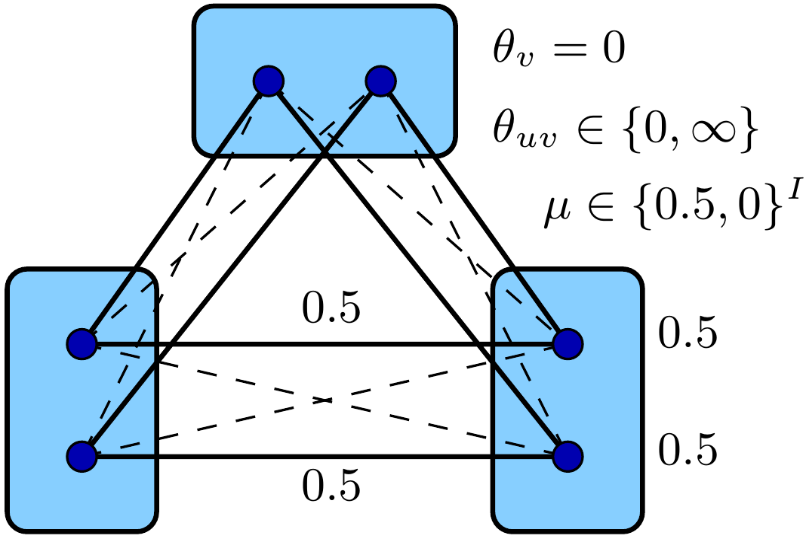
Proposition 5: $\mathcal{M} \subseteq \mathcal{L}$

Check:

$\mu^* \in \mathcal{L}$

$\mu^* \notin \mathcal{M}$

μ - vertex of $\mathcal{L}$



Integer Linear Programs

$$\min_{x \in \mathbb{R}^N} \langle c, x \rangle$$

$$\text{s.t. } Ax \leq b$$

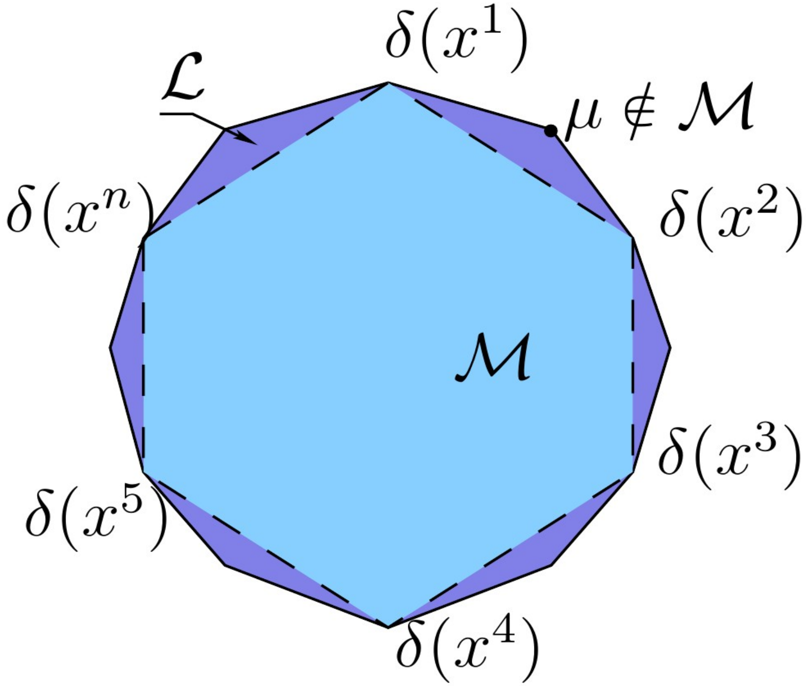
$$x_i \in \{0, 1\}, i \in I \subset \{1, \ldots, N\}$$

$|I|=N$ – integer linear program

$|I|<N$ – mixed integer linear program

NP-hard, standard solvers exist

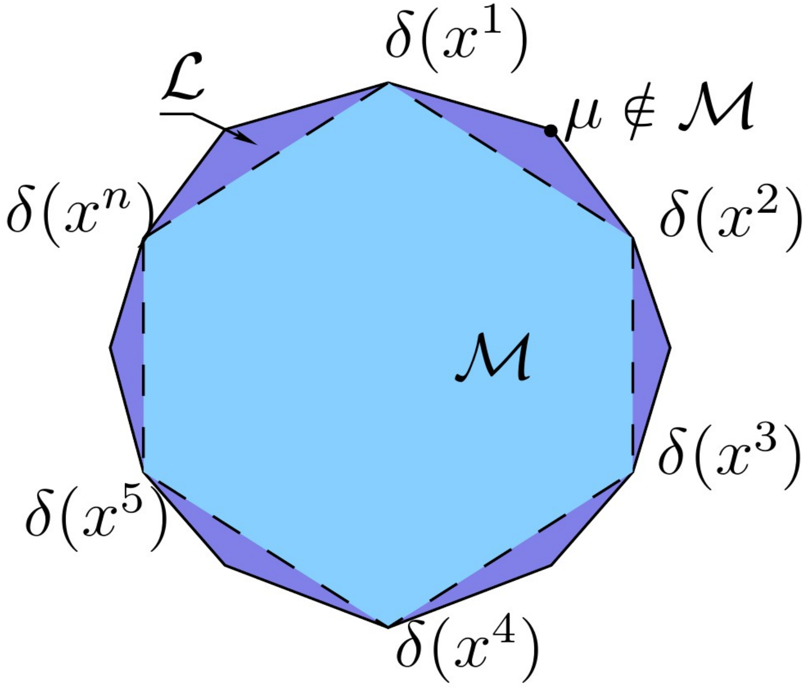
Local (Marginal) Polytope



Corollary 2:

$$\min_{\mu \in \mathcal{L} \cap \{0,1\}^I} \langle \theta, \mu \rangle = \min_{x \in X_V} \langle \theta, \delta(x) \rangle$$

Local (Marginal) Polytope



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?

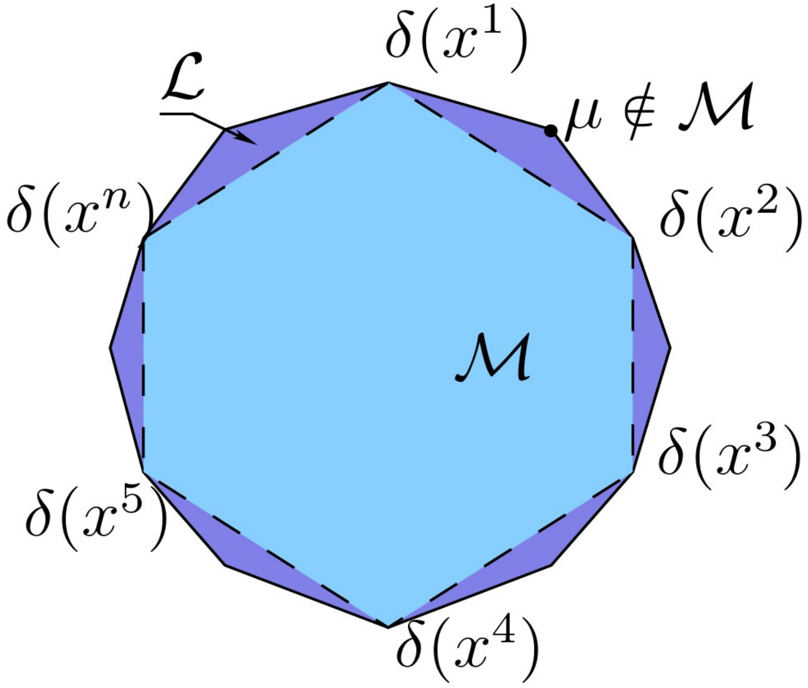
1) $\min_{\mu \in L} \langle \theta, \mu \rangle \leq \min_{x \in X_V} \langle \theta, \delta(x) \rangle$

2) $\ldots \geq \ldots$

3) $\ldots = \ldots$

4) None is correct

Local (Marginal) Polytope



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Local Polytope: Integer/Fractional Solutions

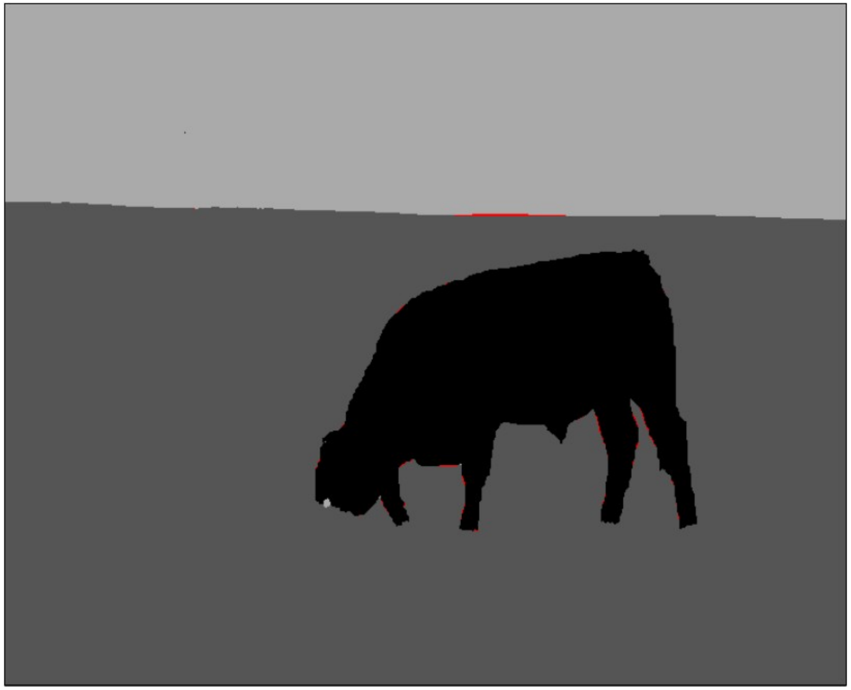
Input Stereopair

Disparities

Pictures and data: Middlebury Benchmark http://vision.middlebury.edu

Local Polytope: Integer/Fractional Solutions

Input Image



Segmentation

Pictures and data: OpenGM Benchmark http://hciweb2.iwr.uni-heidelberg.de/opengm

Local Polytope: Integer/Fractional Solutions

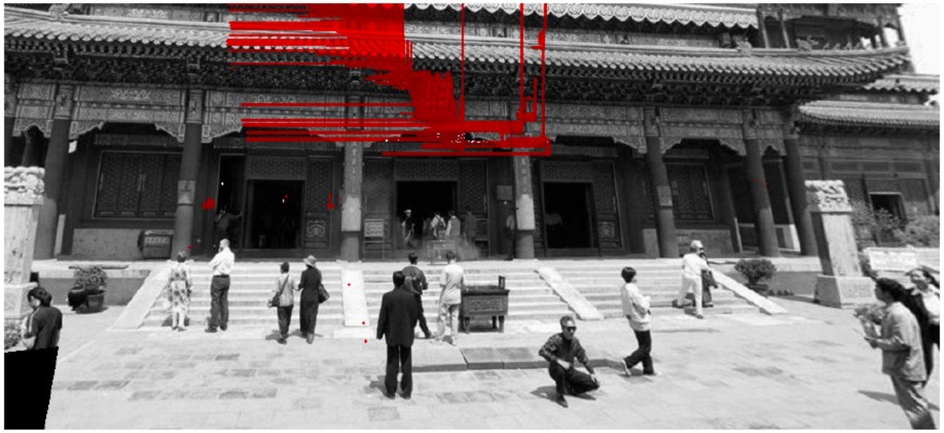
Multiple images

Stitched image

Pictures and data: Middlebury Benchmark http://vision.middlebury.edu

Local Polytope: Integer/Fractional Solutions

Multiple images



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Rounding of a Relaxed Solution

$$x'_v := \arg\max_{s \in X_v} \mu'_v(s), \;\; v \in V$$

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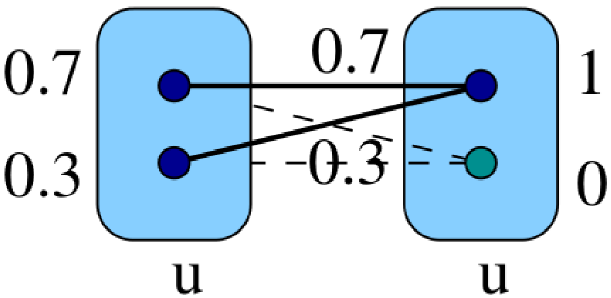
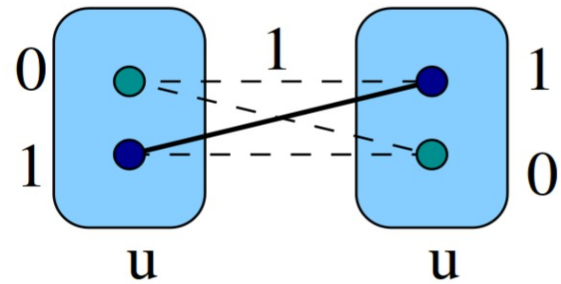
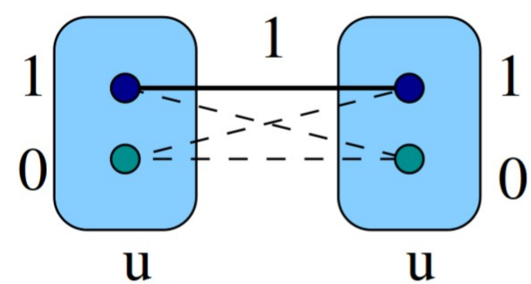


?

1)

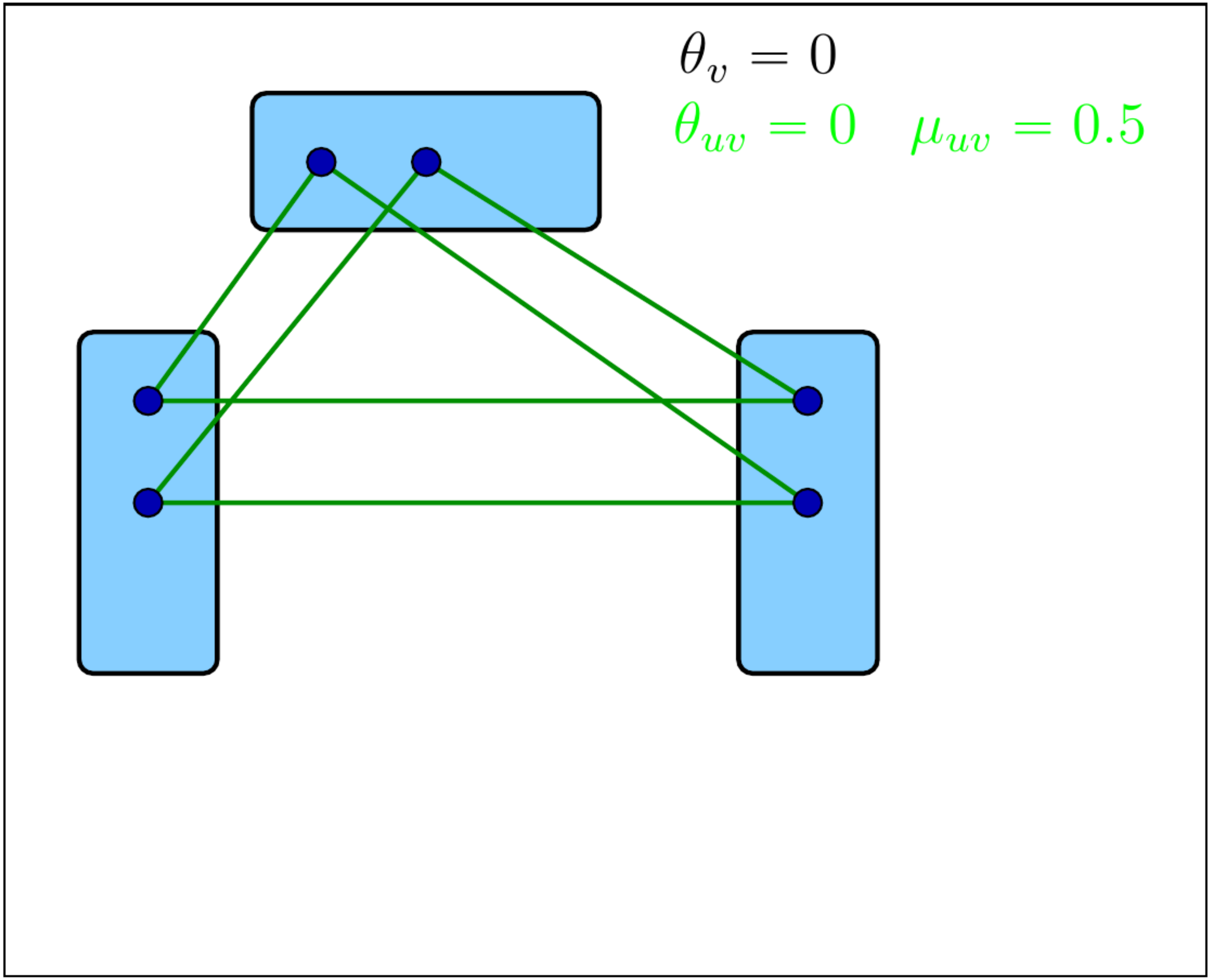


2)

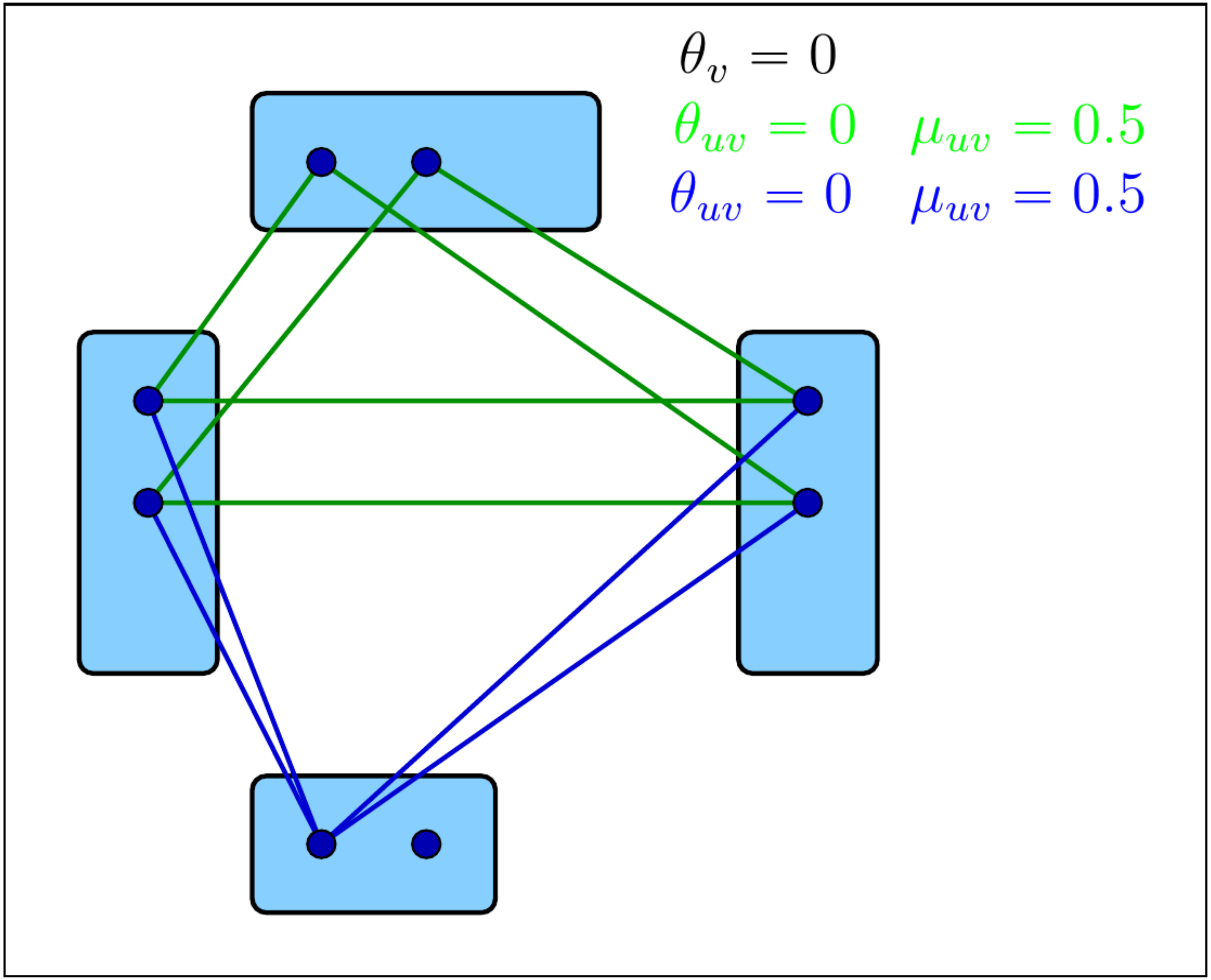


3) None of above

Partial Integrality Does not Mean Partial Optimality



Partial Integrality Does not Mean Partial Optimality



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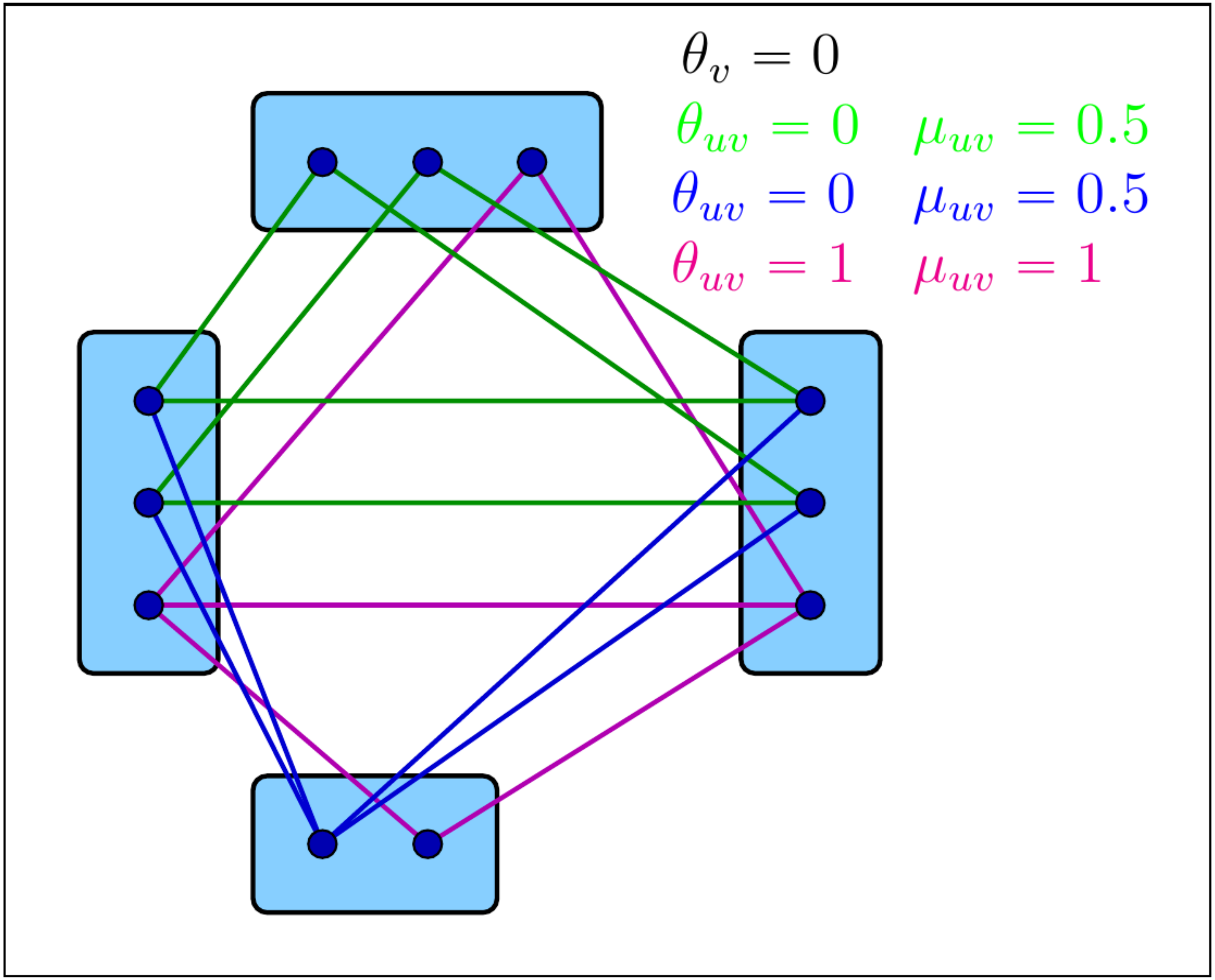
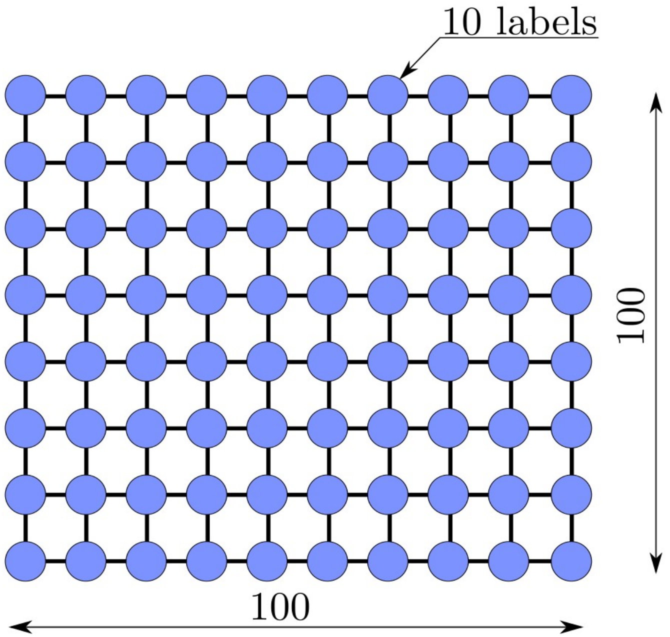


Problem Size



$2 \cdot 10^6$ variables

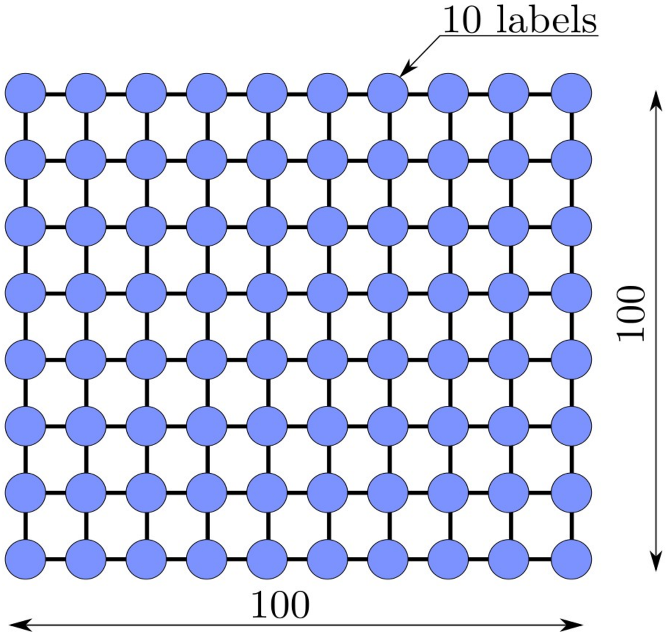
Pascal VOC 2012
semantic segmentation
model $\approx 500 \times 300 \times 21$ labels

$> 10^9$ variables

Number of constraints, order of **?**

1) less
2) 1 000
3) 10 000
4) 100 000
5) 1000 000
6) more

Problem Size



$2 \cdot 10^6$ variables

Pascal VOC 2012
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$> 10^9$ variables

Standard (simplex, interior point) methods do not scale good enough!

Specialized solvers are needed.

Universality of Local Polytope

Daniel Pruša and Tomas Werner, 2014:

Theorem:

"*Any linear program reduces to the local polytope relaxation of certain graphical model with 3 labels in linear time.*"